

Cause and Effect

A cause makes something happen. The effect is what happens next.

Name _____ Date _____

What I am testing

Cause - what I change or do

The one thing I do differently.

My prediction - what I think will happen

I think ... because ...

Effect - what really happened

Was my prediction right?

It is fine to be wrong. Scientists learn from that too.

Cause, Effect and Prediction

Careful: a pattern in your data shows correlation. A cause is something you have to argue for, using a controlled test.

Name _____ Date _____

Question

Proposed cause The factor you will change, and its range.

Prediction and reasoning I predict that ... because ... (name the science that makes you expect it).

Observed effect Data with units. Did the size of the effect change with the size of the cause?

Other possible explanations

What else could have produced this result? Which variables were not controlled?

Conclusion

Cause, Effect and Prediction

Predict first, then test. Writing the prediction down before you start is what makes it a test.

Name _____ Date _____

Question

Cause What you change (the independent variable).

Prediction I predict that ... because ...

Effect observed What actually happened, with measurements.

Explanation Why did it happen? Does the evidence match your prediction?

What would I change next time?
